

TrafficLoc: Localizing Traffic Surveillance Cameras in 3D Scenes

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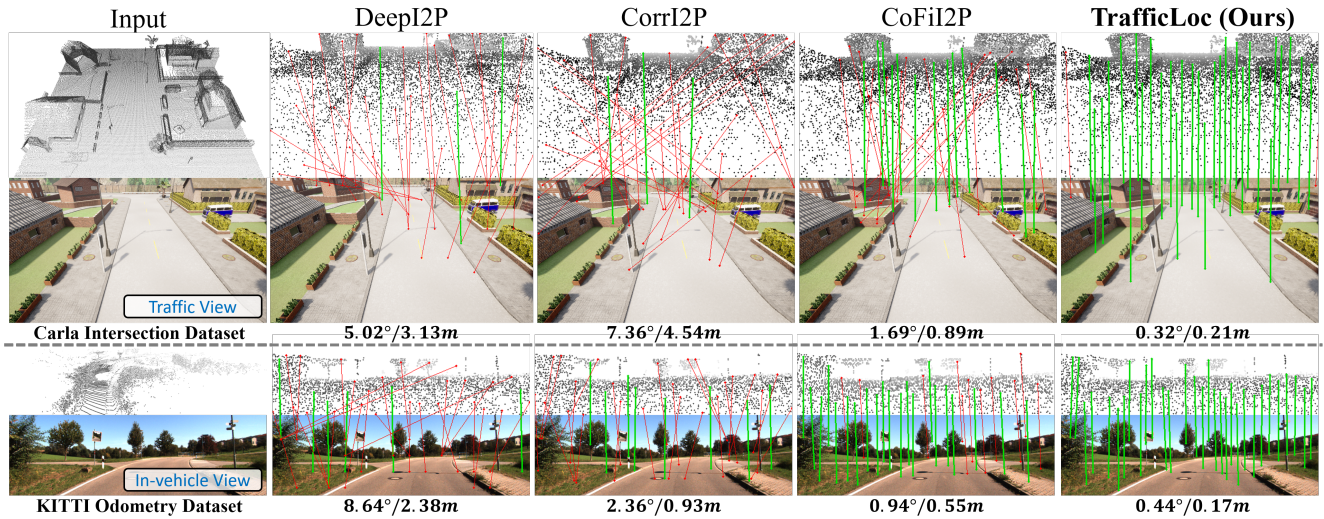


Figure 1. Localization accuracy on the proposed *Carla Intersection* and *KITTI* dataset. The point cloud is projected into a 2D view and shown above the image, with point colors indicating distance. The proposed *TrafficLoc* achieves the best performance, with more correct (green) and fewer incorrect (red) point-to-pixel pairs. The first column presents the input point cloud and input image.

Abstract

We tackle the problem of localizing traffic cameras within a 3D reference map and propose a novel image-to-point cloud registration (I2P) method, *TrafficLoc*, in a coarse-to-fine matching fashion. To overcome the lack of large-scale real-world intersection datasets, we first introduce *Carla Intersection*, a new simulated dataset with 75 urban and rural intersections in *Carla*. We find that current I2P methods struggle with cross-modal matching under large viewpoint differences, especially at traffic intersections. *TrafficLoc* thus employs a novel Geometry-guided Attention Loss (GAL) to focus only on the corresponding geometric regions under different viewpoints during 2D-3D feature fusion. To address feature inconsistency in paired image patch-point groups, we further propose *Inter-intra Contrastive Learning* (ICL) to enhance separating 2D patch / 3D group fea-

tures within each intra-modality and introduce *Dense Training Alignment* (DTA) with *soft-argmax* for improving position regression. Extensive experiments show our *TrafficLoc* greatly improves the performance over the SOTA I2P methods (up to 86%) on *Carla Intersection* and generalizes well to real-world data. *TrafficLoc* also achieves new SOTA performance on *KITTI* and *NuScenes* datasets, demonstrating the superiority across both in-vehicle and traffic cameras. Our project page is publicly available at <https://tum-luk.github.io/projects/trafficloc/>.

1. Introduction

Traffic surveillance cameras, as affordable and easy-to-install roadside sensors in cooperative perception, offer a broad, global perspective on traffic. Integrating this data with onboard sensors enhances situational awareness, supporting applications like early obstacle detection [28, 34, 49] and vehicle localization [42, 54]. Localizing the 6-DoF

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pose (*i.e.* position and orientation) of each traffic camera within a 3D point cloud (e.g., from LiDARs) captured at different locations is essential for cooperative perception.

Recent advances in 3D sensors have enhanced Image-to-Point cloud (I2P) registration methods [18, 20, 35, 51, 53], as LiDAR-acquired point clouds provide accurate and detailed 3D information [47, 48]. However, unlike these I2P methods, the registration of the traffic camera focuses on determining **fixed** camera poses within the point cloud scene. The main challenge of localizing traffic cameras lies in several aspects: 1) Images and 3D reference point clouds are captured at different times and from largely different viewpoints, making it difficult to obtain the precise initial guess required for traditional registration methods [40]. 2) Directly projecting reference point clouds onto the images can lead to a “bleeding problem” [20]. 3) Variable focal length cameras are generally used as traffic cameras for easy installation, causing the intrinsic parameters to be frequently changed during operation [17].

To date, only a few methods have been proposed for localizing traffic cameras in the 3D reference scene. [17] employs manual 2D-3D feature matching followed by optimization using distance transform. [42] uses panoramic images for point cloud reconstruction and then aligns traffic camera images with the resulting point cloud. [40] proposes to automate traffic image-to-point cloud registration by generating synthesized views from point clouds to reduce modality gaps. Although these methods achieve promising results, the requirement for manual intervention or panoramic/rendering image acquisition complicates their deployment. It is therefore important to develop the capability to perform traffic camera localization directly.

However, most of the existing datasets (*i.e.* KITTI [13] and NuScenes [5]) are limited to the in-vehicle cameras, lacking sufficient intersections and traffic cameras. To address this gap, we first introduce *Carla Intersection*, a new intersection dataset using the Carla simulator. *Carla Intersection* provides 75 intersections across 8 worlds, covering both urban and rural landscapes. Furthermore, we propose TrafficLoc, a novel I2P registration method for localizing traffic cameras within a 3D reference map. TrafficLoc follows a coarse-to-fine localization strategy, beginning with image patch-to-point group matching and refining localization through pixel-to-point matching.

We find that simply applying transformers for cross-modal feature fusion in current I2P registration methods [18, 51] struggle with limited geometric awareness and weak robustness to viewpoint variations. To address these challenges, we propose a novel Geometry-guided Feature Fusion (GFF) module, a Transformer-based architecture optimized by a novel Geometry-guided Attention Loss (GAL). GAL directs the model to focus on geometric-related regions during feature fusion, significantly improving per-

formance in scenarios with large viewpoint changes. Another observation is that, when registering image patch and point group features in the coarse matching, the normal contrastive learning in previous I2P methods ignores one fact: *Paired 2D patch-3D point group relationships require separable features within each intra-modality*. To address this issue, we propose a novel Inter-intra Contrastive Learning (ICL) to enhance feature distinctiveness between 2D patches and 3D groups. Moreover, we find that only aligning features with sparsely paired 2D patch-3D groups potentially neglects extra global features. We thus propose a Dense Training Alignment (DTA) to back-propagate the calculated gradients to all image patches using a soft-argmax operator. These operations enable one-stage training to estimate accurate pixel-point correspondences.

To summarize, the main contributions of this work are:

- We set up a new simulated intersection dataset, *Carla Intersection*, to study the traffic camera localization problem in varying environments. *Carla Intersection* includes 75 intersections across 8 worlds, covering both urban and rural landscapes.
- We propose a novel I2P registration method, TrafficLoc, following a coarse-to-fine pipeline. A novel Geometry-guided Attention Loss (GAL) is proposed to direct the model to focus on the geometric-related regions during cross-modality feature fusion.
- We propose a novel Inter-intra Contrastive Learning (ICL) and a Dense Training Alignment (DTA) with a soft-argmax operator to achieve more fine-grained features when aligning each image patch-point group.
- We conduct extensive experiments on the proposed *Carla Intersection*, USTC intersection [40], OpenTraffic-Cam [43], KITTI [13] and NuScenes [5] datasets, showing the proposed TrafficLoc greatly improves over the state-of-the-art methods and generalizes well to real intersection data and in-vehicle camera localization task.

2. Related Work

Uni-modal registration. Image-to-image registration (I2I) methods aim to establish pixel-level correspondences between different images, enabling the estimation of relative transformations via classical algorithms such as PnP [21]. This serves as a fundamental prerequisite for triangulation in Structure-from-Motion (SfM) [1, 16, 30] and SLAM [29]. Traditional methods use handcrafted descriptors such as SIFT [26] or ORB [37] to detect and match keypoints. Recent learning-based methods [8, 10, 36] significantly improve the matching robustness against viewpoint variation. SuperGlue [39] and LightGlue [24] leverage graph neural networks (GNN) and attention mechanisms to refine feature correspondences. Detector-free method LoFTR [41] adopt a coarse-to-fine manner with Transformer to produce semi-dense matches. Efficient LoFTR [46] introduces

a lightweight aggregated attention module for further efficiency improvement. Conventional point cloud-to-point cloud registration (P2P) methods mainly use ICP [3] and its variants [7, 38], which highly rely on initial guess under large relative transformation. Recent learning-based methods [2, 15, 27, 52, 55] extract and match point descriptors, and apply robust estimation methods like RANSAC [12] to optimize rigid transformation. Geotransformer [33] proposes a geometric transformer to learn transformation-invariant point descriptors in the superpoint matching module. EYOC [25] trains a feature extractor using progressively distant point cloud pairs and can adapt to new data distributions without pose labels. While uni-modal registration methods are effective for feature matching within a single modality, they struggle to generalize to cross-modal scenarios due to inherent feature discrepancies.

Image-to-Point cloud registration. To estimate the relative pose between an image and a point cloud, methods like 2D3D-MatchNet [11] and LCD [31] use deep networks to learn descriptors jointly from 2D image patches and 3D point cloud patches. 3DTNet [50] learns 3D local descriptors by integrating 2D and 3D local patches, treating 2D features as auxiliary information. Cattaneo *et al.* [6] establish a shared global feature space between 2D images and 3D point clouds using a teacher-student model. Recent VXP [23] improves the retrieval performance by enforcing local similarities in a self-supervised manner. DeepI2P [22] reformulates cross-matching as a classification task, identifying if a projected point lies within an image frustum. FreeReg [44] employs pretrained diffusion models and monocular depth estimators to unify image and point cloud features, enabling single-modality matching without training. EP2P-Loc [19] performs 2D patch classification for each 3D point in retrieved sub-maps, using positional encoding to determine precise 2D-3D correspondences. CorriI2P [35] directly matches dense per-pixel/per-point features in overlapping areas to establish I2P correspondences, while CoFiI2P [18] and CFI2P [51] employ a coarse-to-fine strategy, integrating high-level correspondences into low-level matching to filter mismatches. Recent VP2P [58] proposes an end-to-end I2P network with a differentiable PnP solver. However, all methods are limited to in-vehicle camera images. In this work, our method generalizes well to car and traffic camera perspectives. Although the high-level strategy is similar to CoFiI2P/CFI2P (and other works that vary this formula), our specific contributions of 3 modules – namely GAL, ICL, and DTA – are what sets it apart.

3. Problem Statement

Let $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ with a resolution of $H \times W$ be a RGB image captured from the traffic cameras. Let $\mathbf{P} \in \mathbb{R}^{N \times 3}$ be a 3D scene point cloud collected by LiDARs at different

locations, where N is the number of points. The camera intrinsic matrix $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ is known or also can be estimated by DUST3R [45] for initialization. Our goal, consistent with previous I2P registration methods [18, 51], is to estimate the 6-DoF relative transformation $\mathbf{T} = [\mathbf{R}|\mathbf{t}]$ between the image $\mathbf{I}$ and point cloud $\mathbf{P}$, including rotation matrix $\mathbf{R} \in \mathbf{SO}(3)$ and translation vector $\mathbf{t} \in \mathbb{R}^3$. To achieve this, we design a neural network to learn a function $\mathcal{F}(\bullet)$ that predicts the set of accurate point-to-pixel correspondences $\mathbf{C}^*$ given $\mathbf{I}$ and $\mathbf{P}$. We then use EPnP-RANSAC [12, 21] to get $\mathbf{T}$ based on the estimated $\mathbf{C}^*$ and $\mathbf{K}$.

4. Our TrafficLoc

Fig. 2 shows our TrafficLoc pipeline. Given a query image and a reference 3D point cloud, we first extract 2D patch features and 3D group features respectively, described in Section 4.1. Next, we fuse the 2D patch and 3D group feature via a carefully designed GFF module, which will be explained in Section 4.2. Section 4.3 describes the coarse matching stage, establishing 2D patch-3D group correspondences. Furthermore, we match each 3D group center feature and the patch features generated from coarse matching in fine matching (Section. 4.4). Lastly, the EPnP-RANSAC module in Section. 4.5 exploits the point-pixel correspondences set to optimize the relative transformation.

4.1. Feature Extraction

Following [19], we explore a dual branch to extract image and point cloud features using Transformer-based encoders.

2D patch descriptors. We first utilize ResNet-18 [14] to extract multi-level features of image I and then use the feature at the coarsest resolution to generate 2D patch descriptors $\mathbf{F}_{\text{patch}} \in \mathbb{R}^{HW/s^2 \times C}$, where s is the resolution of non-overlapping patches and HW/s^2 is the number of image patches. To further enhance the spatial relationships within these image patches, we leverage the pre-trained ViT encoder in DUST3R [45] owing to its strong capability in regressing 3D coordinates. We reduce the original 1024-dimensional output from DUST3R to 256 by employing a multi-layer convolutional network with 1×1 kernels.

3D point group descriptors. We first utilize a standard PointNet [32] to extract point-wise features $\mathbf{F}'_{\mathbf{p}} \in \mathbb{R}^{N \times C'}$ for each point. Then, we conduct Farthest Point Sampling (FPS) to generate M super-points $\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_M\}$ and assign every point in $\mathbf{P}$ to its nearest center in $\mathbf{Q}$, formulating M point groups $G^P = \{G_1^P, G_2^P, \dots, G_M^P\}$ and their associated feature sets $G^{F'} = \{G_1^{F'}, G_2^{F'}, \dots, G_M^{F'}\}$. Following CFI2P [51], we then use the Point Transformer [57] to enhance local geometry features and incorporate global contextual relationships for each point group.

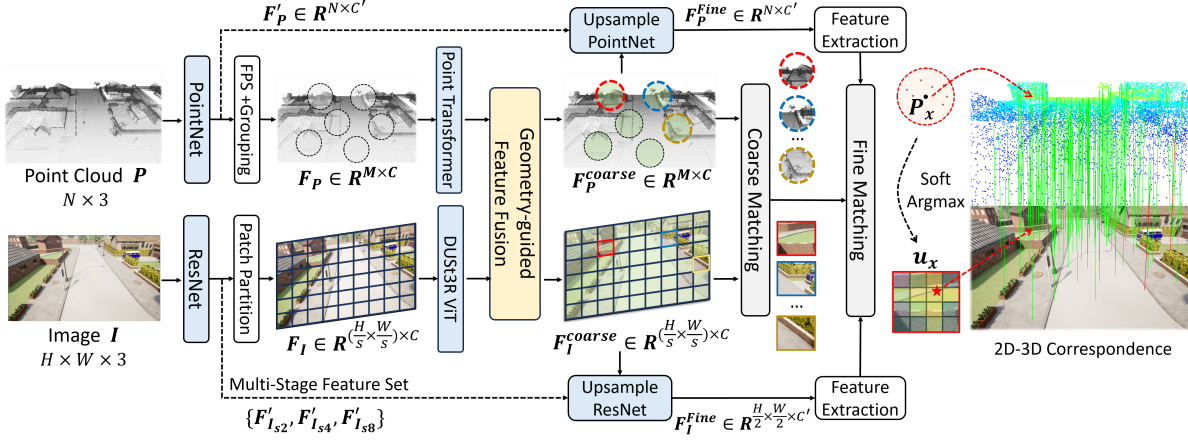


Figure 2. Pipeline of our TrafficLoc. Given a traffic camera image and a 3D scene point cloud collected at different locations, we first extract features at the point group level and image patch level, respectively. We then fuse them using a Geometry-guided Feature Fusion (GFF) module and match them based on similarity rules. Furthermore, we perform fine matching between the point group center and the extracted image window with a soft-argmax operation. Finally, we use EPnP-RANSAC [12, 21] algorithm to get the final camera pose based on the predicted 2D-3D correspondences.

4.2. Geometry-guided Feature Fusion

Although the I2P registration approaches [18, 22] achieved notable fusion success using a Transformer-based architecture, the inherent differences between different modalities and viewpoints are not well-explored. To address this issue, we introduce a novel Geometry-guided Feature Fusion (GFF) module to enhance the network’s robustness to viewpoint variations during cross-modal feature fusion. Specifically, we design a Fusion Transformer architecture guided by a novel geometry-guided attention loss (GAL). Fig. 3 shows the detailed architecture of the GFF module.

Fusion Transformer. Similar to [51], we adopt a Transformer-based architecture with self-attention and cross-attention layers for cross-modal feature fusion. Given the image feature F_I and point cloud feature F_P , we begin by adding sinusoidal positional embeddings to retain the spatial information within both modalities. In the self-attention module, a transformer encoder enhances features in each modality individually using standard scalar dot-product attention. The cross-attention layer is designed to fuse image and point cloud features by applying the attention mechanism across modalities. This design allows for the exchange of geometric and textural information between image and point cloud features, enabling a richer, modality-aware feature representation. More details are in *Supp.*

Geometry-guided Attention Loss (GAL). We find that directly applying Transformers for cross-modal feature fusion suffers from limited geometric awareness and weak robustness to viewpoint variations. We thus propose a novel geometry-guided attention loss that supervises cross-modal attention map during training based on geometric alignment, encouraging features to focus on their geometrically

corresponding parts, as shown in Fig. 3 (right).

Inspired by [4], we apply supervision to the cross-attention layer at the final stage of the Fusion Transformer, leveraging camera projection geometry as guidance. In Image-to-Point cloud (I2P) attention, we encourage the network to focus on relevant 3D point groups for each 2D patch feature I_i . This is achieved by penalizing attention values that have high values outside the target region and encouraging high attention within the desired area. We implement this through a Binary Cross Entropy (BCE) loss on the *raw* cross-attention map ATT_{I2P} :

$$L_{I2P}(i, j) = \text{BCE}(\sigma(ATT_{I2P}(i, j)), \mathbb{1}_{I2P}(i, j))$$

$$ATT_{I2P} = F_I W_Q (F_P W_K)^\top \in R^{HW/s^2 \times M} \quad (1)$$

where σ is a sigmoid function, W_Q, W_K denote the learnable query and key matrices of Transformer, and $\mathbb{1}_{I2P}(i, j)$ is a special indicator function defined as:

$$\mathbb{1}_{I2P}(i, j) = \begin{cases} 1, & \text{if } \text{Rad}(i, j) < \theta_{low} \\ 0, & \text{if } \text{Rad}(i, j) > \theta_{up} \end{cases} \quad (2)$$

The angular radius $\text{Rad}(i, j)$ is given by $\text{Rad}(i, j) = \arccos(\frac{OI_i \cdot OP_j}{\|OI_i\| \|OP_j\|})$. Here, $\mathbb{1}_{I2P}(i, j)$ assigns a value of 1 when the angle between the camera ray OI_i (formed by the 2D patch feature I_i and the camera center O) and the line to the 3D point center P_j is below a threshold θ_{low} . If this angle exceeds an upper threshold θ_{up} , the value is set to 0, indicating that the point is outside the target region. Points with angles between these thresholds are NOT supervised during training, allowing the network to flexibly learn the relationship between attention in these intermediate cases. Similarly, in Point cloud-to-Image (P2I) attention, we encourage each point group to focus on image patches within

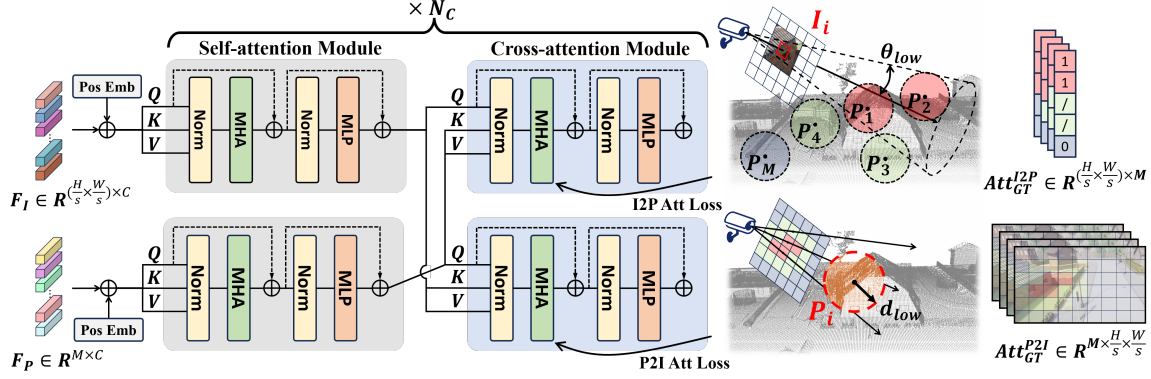


Figure 3. The pipeline of Geometry-guided Feature Fusion (GFF) module. GFF first use N_c layers of self and cross-attention module to enhance the feature across different modalities (left). The proposed Geometry-guided Attention Loss is applied to the cross-attention map of the last fusion layer based on camera projection geometry (right).

its target area of influence:

$$\begin{aligned}
 L_{P2I}(i, j) &= \text{BCE}(\sigma(ATT_{P2I}(i, j)), \mathbb{1}_{P2I}(i, j)) \\
 ATT_{P2I} &= F_P W_Q (F_I W_K)^\top \in \mathbb{R}^{M \times HW/s^2} \\
 \mathbb{1}_{P2I}(i, j) &= \begin{cases} 1 & \text{if } \text{Dist}(P_i, OI_j) < d_{low} \\ 0 & \text{if } \text{Dist}(P_i, OI_j) > d_{up} \end{cases}
 \end{aligned} \quad (3)$$

where $\text{Dist}(P_i, OI_j)$ represents the distance from point center P_i to the camera ray OI_j (formed by the camera center O and the patch I_j) in the 3D space. Since all point groups share a similar receptive field during feature extraction, the distance threshold is set the same for each. This ensures closer objects appear larger while distant ones appear smaller. The final geometry-guided attention loss is:

$$L_{Att} = \sum_{i=1}^{HW/s^2} \sum_{j=1}^M L_{I2P}(i, j) + L_{P2I}(j, i) \quad (4)$$

Note that the relative GT transformation matrix T_{GT} is required only during training. Experiments in Section 5.4 shows the effectiveness of GAL, leading to significant improvements in scenarios with large viewpoint changes.

4.3. Coarse Matching

Here we aim to match the point group with the image patch at the coarse level given the fused 2D patch features $\{F_I^{coarse}\}_{i=1}^{N_p}$ and 3D point group features $\{F_P^{coarse}\}_{i=1}^{N_g}$. Since a monocular camera can only capture a part of the 3D point cloud scene due to the limited field of view (FoV), we first apply a simple super-points filter with binary classification MLP head to predict super-points in or beyond the frustum, supervised by In-frustum Detection Loss L_{det} . For each predicted in-frustum point group P_i , we estimate

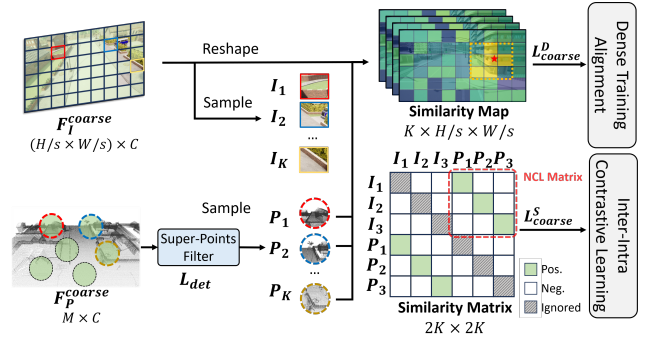


Figure 4. Coarse matching mechanism of TrafficLoc. The positive feature pairs are generated based on ground-truth transformation matrix. The coarse image feature F_I^{coarse} is reshaped to compute its similarity map with each coarse point feature.

its corresponding coarse pixel u_i based on feature similarity to get the predicted coarse correspondence set $\hat{C}_{coarse} = \{(P_i, u_i)\}$, we use cosine similarity $s(\cdot, \cdot)$ to denote the similarity between two features:

$$s(P_i, I_j) = \frac{\langle F_{P_i}^{coarse}, F_{I_j}^{coarse} \rangle}{\|F_{P_i}^{coarse}\| \|F_{I_j}^{coarse}\|} \quad (5)$$

Inter-intra Contrastive Learning (ICL). Unlike normal contrastive learning (NCL) in [18, 51] that solely considers inter-modal pairs, our ICL further incorporates intra-modal pairs (see Figure 4 Similarity Matrix). By constructing negative pairs within the same modality (e.g., I_1 and I_2), the feature distance between different image patches is increased, as well as between different point groups. Therefore, ICL is not only to bring the features of 3D point groups and their corresponding image patches closer together but also to ensure that the 2D patch features within each image and 3D point groups within each point cloud remain as differentiated as possible. This balance enhances more pre-

cise alignment while preserving distinctiveness among local intra-features. Similarity visualizations are in Sec. 5.4.

Our ICL loss is defined as:

$$L_{coarse}^S = \log[1 + \sum_j \exp^{\alpha_p(1-s_p^j+m_p)} \sum_k \exp^{\alpha_n(s_n^k-m_n)}] \quad (6)$$

where s_p^j and s_n^k denote the cosine similarity of positive and negative pairs, respectively. The positive pairs are determined by whether the center of point group P_i is projected into the image patch I_i . α_p and α_n are the adaptive weighting factors for positive and negative pairs, respectively:

$$\begin{aligned} \alpha_p &= \gamma \cdot \max(0, 1 + m_p - s_p^j) \\ \alpha_n &= \gamma \cdot \max(0, s_n^k - m_n) \end{aligned} \quad (7)$$

in which γ is the scale factor and m_p and m_n are positive and negative margin for better similarity separation.

Dense Training Alignment (DTA). Moreover, we find that the proposed ICL can only align features with sparsely sampled image patch-point groups, neglecting additional global features. Inspired by [56], we propose a dense training alignment strategy to back-propagate the calculated gradients to all image patches using a soft-argmax operator. Specifically, for each sampled point group feature P_x , we compute its similarity map $S_x = F_{P_x}^{coarse} F_I^{coarseT} \in \mathbb{R}^{1 \times H/s \times W/s}$ with the target image feature F_I^{coarse} , as shown in Figure 4. Then, we take soft-argmax over the similarity map to compute the predicted pixel position $\hat{u}_x = \text{SoftArgmax}(S_x)$. We penalize the distance between the predicted position $\hat{u}_x$ and the target position u_x with a L2 norm loss, which is defined as:

$$L_{coarse}^D = \sum_x \|\hat{u}_x - u_x\|_2, \quad (8)$$

4.4. Fine Matching

After the coarse matching, we aim to generate point-to-pixel pairs in fine matching. We first generate the image feature $\mathbf{F}_I^{Fine} \in \mathbb{R}^{H/2 \times W/2 \times C'}$ and the point feature $\mathbf{F}_P^{Fine} \in \mathbb{R}^{N \times C'}$ in fine-resolution via applying the two upsample networks, ResNet [14] and PointNet [32], for image and point cloud respectively. We then extract the local feature from the fine-resolution image patch, $\mathbf{F}_{I_x}^{Fine} \in \mathbb{R}^{w \times w \times C'}$, centered on the predicted coarse pixel coordinate $\hat{u}_x$. Additionally, we extract the corresponding feature $\mathbf{F}_{P_x}^{Fine} \in \mathbb{R}^{1 \times C'}$ of the center point P_x within the point group. The fine matching process is defined as:

$$\begin{aligned} \hat{S}_x^{Fine} &= \mathbf{F}_{P_x}^{Fine} \mathbf{F}_{I_x}^{FineT} \in \mathbb{R}^{1 \times w \times w} \\ \hat{u}_x^{Fine} &= \text{SoftArgmax}(\hat{S}_x^{Fine}) \end{aligned} \quad (9)$$

where $\hat{S}_x^{Fine}$ is the fine similarity map between the point center and extracted image patch, and $\hat{u}_x^{Fine}$ is the final predicted 2D pixel corresponding to 3D point P_x .

4.5. Pose Estimation

Following [18], we use the EPnP-RANSAC [12, 21] algorithm to filter out incorrect pixel-point pairs and estimate the relative pose of the camera based on the set of predicted pixel-point correspondence after the fine matching.

4.6. Loss Function

Our training loss includes four components:

Geometry-guided Attention Loss. We use a Binary Cross Entropy (BCE) loss L_{Att} to supervise the cross-attention layers at the final stage of the Fusion Transformer during cross-modal feature fusion (See Sec. 4.2).

In-frustum Detection Loss. We use a standard binary cross-entropy (BCE) loss L_{det} (See Fig. 4) to supervise the in-frustum super-points classification.

Coarse Matching Loss. In coarse matching, we first sample κ pairs from the ground-truth 2D-3D corresponding set $C_{coarse}^* = \{(P_x, I_x) | I_x = \mathcal{F}(T_{GT}P_x)\}$, where T_{GT} is the ground-truth transformation matrix from point cloud coordinate system to image frustum coordinate system, and $\mathcal{F}$ denotes the mapping function that convert points from camera frustum to image patch position. We determine the negative pairs by checking whether the distance between the location of I_x and the projection of P_x on the image is larger than a threshold r or not. The process is the same within each individual modality. Then, we combine the Inter-intra Contrastive Learning loss and the loss in Dense Training Alignment to form our coarse matching loss $L_{coarse} = L_{coarse}^S + L_{coarse}^D$.

Fine Matching Loss. Since the fine image patch has a small region, we utilize Cross Entropy (CE) loss L_{fine}^S to sparsely supervise the fine matching process and apply dense L2 norm loss L_{fine}^D for constructing fine matching loss $L_{fine} = L_{fine}^S + L_{fine}^D$:

$$\begin{aligned} L_{fine}^S &= -\frac{1}{\kappa} \sum_{x=1}^{\kappa} \sum_{c=1}^{w^2} S_{x,c}^{Fine} \log(\text{softmax}(\hat{S}_{x,c}^{Fine})) \\ L_{fine}^D &= \sum_x \|\hat{u}_x^{Fine} - u_x^{Fine}\|_2 \end{aligned} \quad (10)$$

where similarity value $S_{x,c}^{Fine}$ equals to 1 when the 3D point P_x is corresponded with the c_{th} pixel in the flattened image patch, else 0. κ is the number of sampled 2D-3D pairs, w is the side length of image patch and u_x^{Fine} denotes the GT corresponded 2D pixel. To avoid overfitting, we randomly shift the centered position of the extracted fine patch by up to $\pm w/2$ pixels, preventing the ground-truth pixel from always being at the center of the patch.

Overall, our loss function is:

$$L = \lambda_1 L_{Att} + \lambda_2 L_{det} + \lambda_3 L_{coarse} + \lambda_4 L_{fine} \quad (11)$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ regulate the losses' contributions.

5. Experiments

5.1. Proposed Carla Intersection Dataset

The proposed *Carla Intersection* dataset comprises 75 intersections across 8 worlds within the Carla [9] simulation environment, encompassing urban and rural landscapes. We use on-board LiDAR sensor to capture point cloud scans, which are then accumulated and downsampled to get the 3D point cloud of the intersection. For each intersection, we captured 768 training images and 288 testing images with known 6-DoF pose at a resolution of 1920x1080 pixel and a horizontal field of view (FOV) of 90°. In consideration of real-world traffic surveillance camera installations, our image collection spans heights from 6 to 8 meters, with camera pitch angles from 15 to 30 degrees. This setup reflects typical positioning to capture optimal traffic views under varied monitoring conditions.

To access the model’s generalization capability, we trained on 67 intersections from worlds *Town01* to *Town07*, and tested on one unseen intersection from each of the 8 worlds. We divided the testing data into three sets:

Test_{T1-T7} contains 7 unseen intersections from world *Town01* to *Town07*, with 288 testing images each, matching the training pitch angles (15 or 30 degree).

Test_{T1-T7hard} contains 7 same intersection scenes as in **Test_{T1-T7}**, with images at pitch angles of 20 or 25 degrees, to evaluate robustness against viewpoint variations.

Test_{T10} contains 1 unseen intersection scene from unseen world *Town10*. This split sets a high standard for evaluating the model’s generalization ability, assessing its performance in unseen urban styles and intersections.

5.2. Experimental Setup

Datasets. We conduct experiments on the *Carla Intersection* dataset. We also evaluate TrafficLoc on the real USTC intersection [40] and OpenTrafficCam [43] datasets, and two in-vehicle camera benchmarks, KITTI Odometry [13] and Nuscenes [5]. For fair comparisons, all models are trained on the same intersections in *Carla Intersection* dataset and evaluated on the same testing intersections. Moreover, to show the generalization ability on the in-vehicle datasets, we used the same training and evaluation pairs of image and point cloud data on KITTI and Nuscenes, following previous I2P registration methods [22, 35].

Implementation Details. The training details and others are in the Supplementary Materials.

Evaluation Metrics. Following previous works [18, 22, 35], we evaluate the localization performance with relative rotation error (RRE), relative translation error (RTE) and registration recall (RR). RRE and RTE are defined as:

$$\text{RRE} = \sum_{i=1}^3 |\mathbf{r}(i)|, \quad \text{RTE} = \|\mathbf{t}_{gt} - \mathbf{t}_{pred}\| \quad (12)$$

where $\mathbf{r}$ is the Euler angle vector of $\mathbf{R}_{gt}^{-1}\mathbf{R}_{pred}$, $\mathbf{R}_{gt}$ and $\mathbf{t}_{gt}$ are the ground-truth rotation and translation matrix, $\mathbf{R}_{pred}$ and $\mathbf{t}_{pred}$ represent the estimated rotation and translation matrix. RR denotes the fraction of successful registrations among the test dataset. A registration is considered as successful when the RRE is smaller than $\tau_r = 10^\circ$ and the RTE is smaller than $\tau_t = 5m$.

5.3. Evaluation Results

We first evaluate TrafficLoc on our *Carla Intersection* dataset and compare it with other baseline methods. Table 1 summarizes the results. Our method outperforms all baseline methods by a large margin in all three test splits. This indicates that TrafficLoc is robust to large viewpoint changes and has great generalization ability on **unseen** traffic scenarios. Specifically, the RRE reduces **85%**, **66%**, **86%** and the RTE reduces **82%**, **78%**, **64%** compared to the previous state-of-the-art CoFiI2P [18] in three test splits respectively. Even in entirely unseen city-style environments and unseen scenes (**Test_{T10}**), our model maintains robust localization capability, while other baseline methods fail. Additionally, our model achieves high accuracy while maintaining efficient inference time. When intrinsic parameters K are unknown, we use intrinsics predicted by DUST3R [45] to initialize the intrinsics for EPnP-RANSAC [12, 21], which slightly decreases performance and inference speed but still yields strong results. The result of VP2P [58] on *Carla Intersection* dataset and its RR on Nuscenes are unavailable since their code is not provided.

Besides *Carla Intersection* dataset, we evaluate our TrafficLoc on KITTI and Nuscenes benchmarks. TrafficLoc achieves the best performance on both datasets, achieving **24%** improvement in RRE and **34%** in RTE compared to the previous state-of-the-art CoFiI2P [18] and $< 1^\circ$ RRE first-time on KITTI, indicating its strong ability also on in-vehicle view cases.

Evaluation on real-world datasets. To test the Sim2Real generalizability of our TrafficLoc, we evaluate it on the real-world intersections from the USTC Intersection [40] and OpenTrafficCam [43] datasets, after only training on the synthetic *Carla Intersection*. The test intersections are totally unseen, and the traffic cameras are uncalibrated. Our model achieves great Sim2Real capability in real-world traffic scenarios. Detailed results are given in the Supplementary Materials.

5.4. Ablation Study

We evaluate the effectiveness of different proposed components in our TrafficLoc on the *Carla Intersection* dataset **Test_{T1-T7}** here. Additional ablation studies and discussions, particularly for ablations on the KITTI Odometry dataset, are presented in the **Supplementary Materials**.

Geometry-guided Attention Loss (GAL). Fig. 6 high-

	Test _{T1-T7}		Test _{T1-T7hard}		Test _{T10}		KITTI Odometry			Nuscenes			Runtime
	RRE(°)	RTE(m)	RRE(°)	RTE(m)	RRE(°)	RTE(m)	RRE(°)	RTE(m)	RR(%)	RRE(°)	RTE(m)	RR(%)	
DeepI2P-Cls [22]	9.02	6.31	9.10	6.12	18.30	11.46	5.88	1.13	80.18	7.37	2.22	62.67	0.23
DeepI2P-2D [22]	20.31	7.01	33.43	7.93	49.93	17.09	3.87	1.42	74.50	3.02	1.95	92.53	8.78
DeepI2P-3D [22]	17.97	7.29	34.86	9.49	43.11	17.41	6.08	1.21	38.34	7.06	1.73	18.82	18.62
CorrI2P [35]	20.08	12.63	27.95	13.97	32.23	14.45	2.72	0.90	92.19	2.31	1.7	93.87	1.33
VP2P [58]	/	/	/	/	/	/	2.39	0.59	95.07	2.15	0.89	/	0.76
CFI2P [51]	4.46	<u>1.92</u>	8.56	<u>2.48</u>	<u>10.81</u>	<u>7.15</u>	1.38	0.54	99.44	<u>1.47</u>	1.09	<u>99.23</u>	<u>0.33</u>
CoFiI2P [18]	<u>4.24</u>	2.82	<u>7.87</u>	5.34	17.78	7.43	<u>1.14</u>	<u>0.29</u>	100.00	1.48	<u>0.87</u>	98.67	0.64
Ours with K_{Pred}	2.04	1.72	4.56	2.19	4.61	4.80	/	/	/	/	/	/	1.74
Ours with K_{GT}	0.66	0.51	2.64	1.13	2.53	2.69	0.87	0.19	100.00	1.38	0.78	99.45	0.85

Table 1. Quantitative localization results on the proposed *Carla Intersection*, *KITTI* [13] and *Nuscenes* [5] datasets. We report median RRE and median RTE for *Carla Intersection* and mean RRE and mean RTE for *KITTI* and *Nuscenes* following previous Image-to-Point cloud registration methods [18, 22, 35]. Our model achieves the best performance on all datasets, especially on unseen scenarios.

lights the advanced geometry-awareness ability of the proposed GAL. With the use of GAL, the P2I attention map of point group P_{3D} tends to concentrate more on the image region where the point group is projected, while the I2P attention map for patch I_{2D} assigns greater weights to the area traversed by the camera ray of this patch. As shown in Table 2 (last two rows), with the proposed L_{att} , our model brings notable improvements (**17.7%**) in the RTE metric.

Inter-intra Contrastive Learning (ICL). Fig. 5 (c) and (d) vividly show the advantages of the proposed ICL. When using existing normal contrastive loss (NCL), the similarity map exhibits generally high values throughout. With the proposed ICL, the distinction within the similarity map increases, demonstrating our TrafficLoc can achieve a more pronounced distributional difference between features within the same modality. Quantitative results in Table 2 (first two rows) demonstrate we achieve **9.8%** improvements in RTE with the proposed ICL compared to using previous NCL.

Dense Training Alignment (DTA). From Fig. 5 (d), we can see that multiple peaks (red regions) still remain due to the sparse supervision during training. With the proposed DTA, the similarity map in Fig. 5 (e) only has a single peak region, showing strong robustness in feature matching. Table 2 (second and third rows) sees an improvement of **20.5%** and **16.2%** in RRE and RTE after using the proposed DTA. This is due to our DTA having global supervision across the image patch-point group matching during training.

	CM	FM	GAL	RRE (°)	RTE (m)
Baseline	NCL	/	/	1.53	0.82
	ICL	/	/	1.27	0.74
	ICL + DTA	/	/	1.01	0.62
	ICL + DTA	✓	/	0.84	0.62
Ours	ICL + DTA	✓	✓	0.66	0.51

Table 2. Ablation Study of our TrafficLoc on the *Carla Intersection* Test_{T1-T7}. CM denotes Coarse Matching and FM denotes Fine Matching. “GAL” indicates the proposed Geometry-guided Attention Loss. “NCL” means using normal contrastive learning, while “ICL” means using the proposed Inter-intra Contrastive Learning. “DTA” indicates using the proposed Dense Training Alignment.

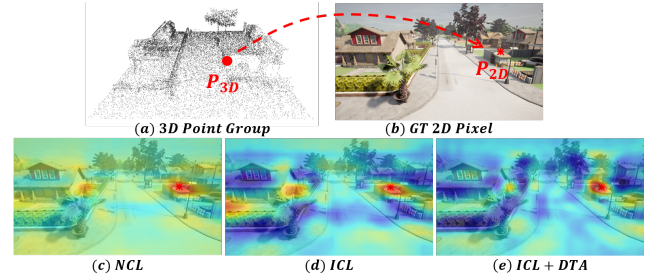


Figure 5. Visualization result of using different loss function. (a), (b) denote the point group center P_{3D} and its corresponding pixel P_{2D} . (c), (d), (e) show the similarity map between point group and image feature. Blue means low similarity and red means high.

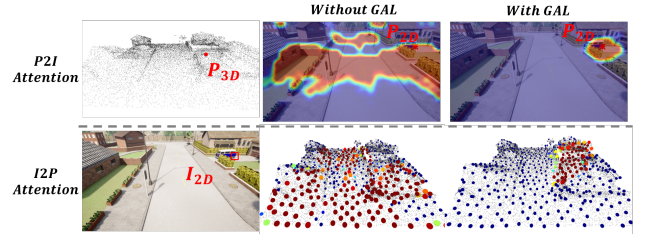


Figure 6. Visualization result of P2I and I2P attention map when using Geometry-guided Attention Loss L_{att} or not. Red color indicates high attention value and blue means low value.

6. Conclusion

In this work we focus on the under-explored problem of traffic camera localization, which is an important capability for fully-integrated spatial awareness among city-scale camera networks and vehicles. Such large-scale sensor fusion has the potential to enable more robustness, going beyond the limitations of a single vehicle’s point of view. We proposed a novel method, TrafficLoc, which we show to be effective. To facilitate training and evaluation we propose the novel *Carla Intersection* dataset, focusing on the case of intersections, which is a common placement for traffic cameras and a focal point for traffic safety. We hope that this dataset will facilitate more research into integrated camera networks for robust, cooperative perception.

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